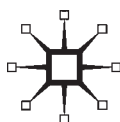


The Scientific Revolution and the Origins of Modern Science

Third edition

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1 The Scientific Revolution and the Historiography of Science

The Scientific Revolution is the name given by historians of science to the period in European history when, arguably, the conceptual, methodological and institutional foundations of modern science were first established. The precise period in question varies from historian to historian, but the main focus is usually held to be the seventeenth century, with varying periods of scene-setting in the sixteenth and consolidation in the eighteenth. Similarly, the precise nature of the Revolution, its origins, causes, battlegrounds and results vary markedly from author to author. Such flexibility of interpretation clearly indicates that the Scientific Revolution is primarily a historian's conceptual category. But the fact that the notion of the Scientific Revolution is a term of convenience for historians does not mean that it is merely a figment of their imaginations with no basis in historical reality.

Certainly, knowledge of the natural world can easily be seen to have been very different in 1700 from the way it was in 1500. Undoubtedly, during this period, highly significant and far-reaching changes were brought about in all aspects of European culture concerned with the nature of the physical world and how it should be studied, analysed and represented, and many of these developments continue to play a significant part in modern science. An entirely contemplative natural philosophy (knowledge for its own sake), based almost entirely on the teachings of the Ancient philosopher Aristotle, was replaced by a general belief that natural knowledge should be put to practical use for the benefit of mankind [117; 150; 68]. Reliance on Ancient authority as the supreme source of knowledge was replaced by a belief that knowledge of the natural world can only be acquired by studying the world at first hand, either by detailed observation or even by specifically contrived investigatory experiments [69; 131; 318]. Where mathematics was once seen as a completely man-made art, and mathematical analysis as too abstract to have any relevance to an understanding of the natural world, mathematics increasingly came to be seen as an indispensable way of revealing the operations of natural phenomena [321; 68; 319; 105]. After centuries in

which 'laws of nature' were simply a vague metaphor, specific and precise laws became codified for the first time [151; 180]. After centuries in which the natural and the man-made or artificial were seen as categorically distinct, the natural world increasingly came to be understood by analogy with artificial processes, culminating, by the end of the seventeenth century, in the view that the natural world was best understood in terms of analogies with mechanical actions such as impact, and physical engagement [12; 69; 76; 111; 131; 318]. Consequently, a belief that all bodies below the moon were made up of combinations of the four elements of earth, water, air and fire, and that above the moon all was composed of an unchanging ethereal fifth element, was replaced by a belief that there was no disjunction between the heavens and the earth, and that everything (both above and below the moon) was composed of invisibly small particles (interacting by impact and intermeshing), and all was subject to the same universal laws of nature [21; 69; 111; 303; 318]. Accordingly, an earth-centred finite cosmos had been replaced by a belief in an infinite universe in which the sun was just one among many stars and the earth was merely one of the planets circling that star [184; 185; 186; 148]. Numerous discoveries in human anatomy led to the discovery of the circulation of the blood and therefore to a complete reassessment of physiology [61; 63; 101; 133; 102; 103]. Sexual reproduction was understood to depend upon the union of sperm and egg, and this was even recognized in plants [1; 101]. These are only some of the most far-reaching changes.

It is perfectly clear, also, that many of the leading innovators in the period saw themselves as doing something new. Indeed, the word 'new' occurs time and again in the titles of many of the books written by the leading figures in the Revolution. Francis Bacon (1561–1626) announced his intention to replace the doctrines of Aristotle's *Organon* in his *New Organon* (1620), Galileo (1564–1642) wrote a book about *Two New Sciences* (1638) and Johannes Kepler (1571–1630), who had earlier noted that the Paracelsians had invented a new medicine and that the Copernicans had invented a new astronomy, presented a book on his own discoveries which he also saw as *A New Astronomy* (1609). There were many similar examples. The concept of the Scientific Revolution can be seen, therefore, to refer to a very real process of fundamental change. If we wish to understand the nature and causes of these changes, we must try to pinpoint the fundamental issues for past thinkers, their most significant switches in ways of thinking, the clearest shifts in their social organization, the most far-reaching changes in their scientific practice, and the implications of the most significant discoveries and inventions. We need not detain ourselves,

however, in discussions about the correct starting date, about precisely what kind of revolution it was, or about the best way of defining revolutionary change in science. To do so is mistakenly to regard what is nothing more than a convenient term of reference for a wide range of major social and intellectual changes as though it somehow grasps the putative essence of those changes [6; 45; 237; 271; 69; 33].

The reification of the Revolution, as a revolution, has, however, given rise to one important historiographical debate; one which continues to be disputed. A number of historians have argued that the very concept of a revolution in early modern science, with its implication of a radical break with the past, is misplaced or misconceived. The issue depends, of course, entirely upon whatever criteria are used to circumscribe the debate [58; 57; 244; 124; 192]. The current consensus seems to be that the 'continuist' view of scientific development has been overstated in the past, but remains valuable for pointing to the many and various antecedents of later developments during the medieval period [192; 76; 33]. As a number of recent surveys of medieval science have made abundantly clear, medieval natural philosophy provided the foundations upon which the Scientific Revolution was built. But these recent works are not strictly continuist in their historiography because they do acknowledge that something significantly different did take place in the period of the Scientific Revolution. There was a change in the edifice of European knowledge which enables us to see where the foundations end and the superstructure begins. The fact remains, however, that where once the Middle Ages could be presented as a period of scientific sterility and stagnation, thanks to the excellent work of continuist historians, we can now see the undeniable achievements of medieval thinkers, particularly in the fields of astronomy and cosmology, optics, kinematics and other mathematical sciences, as well as in the development of the notion of natural laws and of the experimental method [123; 124; 57; 58; 192; 244; 76; 151; 220; 221].

Moreover, continuist historiography has played an important part in making historians of science aware of the dangers of what is called 'whiggism'. There is a tendency in the history of science to look back with hindsight about what is known to be important later. To judge the past in terms of the present is to be whiggish. In the early decades of the formation of the discipline it was common for a historian of science to pick out from, say, Galileo's work, or Kepler's, those features which were, or could most easily be made to look like, direct anticipations of currently held science. The resulting history was often a lamentable distortion of the way things were. But the very notion of the Scientific Revolu-

tion, it's easy to see, has something rather whiggish built into it. The science of that era was revolutionary because, unlike previous science, it was like our own, or so we think. It's almost as if what we want to say is not just, here are the origins of modern science, but here is the beginning of current science.

There is a sense in which this kind of whiggism still thrives in the history of science. The *raison d'être* of the history of science is, essentially, to try to understand why and how science became such a dominant presence in our culture [183; 171; 48; 59; 158]. As such, all our history is directed towards the present. So, although the vigorous repudiation of whiggism has now become a shibboleth which must be uttered to gain entry into the ranks of serious scholars, whiggism lurks within all of us [130]. The distinguished intellectual historian, Richard H. Popkin, once wittily announced that he intended to study the reasons why 'one of the greatest anti-Trinitarian theologians of the seventeenth century', Isaac Newton (1642–1727), should take time off to write works on natural science [236]. We assume this is meant to be witty because we find it impossible to take seriously the suggestion that Newton's historical significance derives from his standing as a theologian. To this extent I have to confess to whiggism. I believe that we study Newton because he made such exceptional contributions to our scientific culture. Nothing else about this fascinating man is quite so interesting as that.

Continuism can be seen, however, at least to some extent, as an antidote to whiggish tendencies because it tends to be backward-looking, rather than inherently forward-looking. The attempt to see Galileo as a latter-day *impetus* theorist is intrinsically less whiggish than presenting him as one who prefigured Newtonian inertia, and should lead us closer to Herbert Butterfield's suggestion (before he himself became a Whig historian by writing a book about the Scientific Revolution! [130: 58; 33]) that we should attempt 'to see life with the eyes of another century than our own' [quoted in 130: 48]. Virtually all historians of science now try to avoid overt forms of whiggism, revolutionists and continuists alike, but it seems safe to say that historians of the Middle Ages were among the first to show the way [57; 58; 76; 124; 192].

Another indicator that the concept of the Scientific Revolution is inherently whiggish is the very word 'scientific'. Our present use of the word 'science' was first coined in the nineteenth century and, strictly speaking, there was no such thing as 'science' in our sense in the early modern period. To talk as though there was, as I have been doing, is an obviously whiggish distortion. Part of our aim, in looking at the historical development of what we think of as science, should be to understand how the very concept 'science'

arose; we simply beg the question if we talk about 'science' as though it always existed.

So, if there was no 'science' at the time of the Scientific Revolution, what was there? There was something called 'natural philosophy', which aimed to describe and explain the entire system of the world [124; 21]. There were a series of technically developed disciplinary traditions, either mathematically based like astronomy [82; 186; 188; 79; 148], optics [250; 57; 198], mechanics [76; 181; 190; 12; 13; 15] and what was called music, but which we would see as a rather more mathematical study based on principles of ratios and other aspects of proportion [43; 47; 122; 198]; or medically based like anatomy [34; 102; 61; 63], physiology [133; 308; 25] and pharmacology or the study of *materia medica*, those things from which medicines could be made [7; 25; 49; 73]. And finally, there were a range of practical arts like navigation, cartography, fortification and other military arts, mining, metallurgy and surgery [271: 225–7; 15; 188; 181; 190]. The relationship between these technical disciplines and natural philosophy requires careful elucidation and work in this area is continuing.

Some of the most exciting research in the history of science has been concerned to show how changing interactions between the specialist disciplines and natural philosophy, through practitioners in either or both camps, have given rise not only to new developments in knowledge and practice, but also to something which looks closer, or more directly related, to our present-day demarcation of scientific disciplines. Galileo's endeavour to bring together kinematics and natural philosophy resulted in what he called the 'new science of motion', which historians still regard as an influential step towards subsequent theories [283]. Similarly, the new and highly influential natural philosophy of René Descartes (1596–1650), the mechanical philosophy, was forged out of his attempts to base natural philosophy upon the certainties of geometrical reasoning [115]; and Newton's new natural philosophy was based, as the title of his book proclaimed, on mathematical principles [106]. The development of atomistic theories of matter grew at least partly out of the efforts of medically trained natural philosophers to extend Aristotle's natural philosophy to account for the empirical knowledge of chemists [220; 221; 216; 86; 206; 303]. The new experimental philosophy, developed in late seventeenth-century England by Robert Boyle (1627–91) and others, was intended to demarcate new discipline boundaries around correct natural philosophy, and to exclude what had previously been regarded as correct practice [279].

A simple but essentially accurate way of summing up what took place in the Scientific Revolution, then, is to say that the natural

philosophy of the Middle Ages, which had tended to remain aloof from mathematical and more pragmatic or experiential arts and sciences, became amalgamated with these other approaches to the analysis of nature, to give rise to something much closer to our notion of science. The Scientific Revolution should not be seen as a revolution *in* science, because there was nothing like our notion of science until it began to be forged in the Scientific Revolution out of previously distinct elements [6].

It should be clear from this that it is by no means ideal to use the term 'natural philosophy' instead of 'science' when dealing with the early modern period. The terms are by no means equivalent. One of the revolutionary things about the Scientific Revolution is precisely the fact that, throughout the period, natural philosophy was being changed beyond all recognition, and approaching closer to our concept of science [21; 111]. Even so, the term 'natural philosophy' was the one which was most used at the time to refer to an understanding of the physical world, and continued to be until the nineteenth century (when the word 'science' acquired its present meaning). Accordingly, I will use 'natural philosophy' and 'science' quite interchangeably, meaning in both cases nothing more than the endeavour to understand, describe or explain the workings of the physical world (I will also use the adjectival forms, 'natural philosophical' and 'scientific' in the corresponding way). I hope neither anachronism will prove too distracting.

It is possible to acknowledge a whiggishness in one's reasons for looking at the history of science without, however, allowing whiggism to intrude into our historical narratives. Rather than imposing our own views, our aim as historians should be to strive for as full an understanding as possible of the contemporary context. For example, if we wish to understand the contemporary response to a little book like Galileo's *Siderius Nuncius* (*Starry Messenger*, 1610), in which Galileo presented the discoveries he had made by turning the newly invented telescope to the night sky, it is obvious that we cannot simply read Galileo's text [108; 292]. Nor will it be enough to familiarize ourselves with the technical astronomy and cosmology of Galileo's time. It is well known, for example, that some of Galileo's contemporaries refused to look through his telescope. Why did they respond that way? Obviously not because of any astronomical technicality [301]. Part of the answer is that magicians, and even common conjurors, used combinations of mirrors and lenses to fool people; and as artificially constructed objects, it was by no means obvious to contemporaries that telescopes could provide reliable knowledge about what was natural [15: 685; 136]. An understanding of the implications of what Galileo wrote is also

relevant. A contemporary reader would have responded differently to a work which was likely to have no impact beyond its immediate area, than to one, like Galileo's, which could be seen to run counter not only to current astronomy and cosmology, but also to the wider natural philosophy, and to religious belief [20; 282; 284; 196; 93; 148]. A really full account would also require some knowledge of contemporary understanding of who Galileo was: his reputation, his presumed or perceived motivation, and whether he could be held to speak disinterestedly, and so forth [19; cf., for similar considerations about other thinkers, 17; 195; 276; 278; 279]. Of course, there are no definite limits to an enterprise of this kind, which is why no single historian can have the last word on any given topic. It is always possible to point to something else in the subject's milieu which may be relevant to our attempt to reconstruct the past.

A striving for an ever richer contextualization can be seen, then, as the driving force in the current historiography of science. This contextualism can be seen as the outcome of an eclecticism which combined what used to be seen as opposite approaches to the history of science. The discipline of the history of science used to be riven by warfare between internalists and externalists (*c.* 1930–59). The internalists were supposed to have believed that science, or possibly an individual sub-discipline within science, was a system of thought which was self-contained, self-regulating, and developed in accordance with its own internal logic. The externalist, on the other hand, was supposed to believe that the development of science was determined by the sociopolitical or socioeconomic context from which it emerged. In fact, neither position seems to have been properly established as valid or viable [277: 345–51], and it wasn't long before a professed eclectic approach became all the rage (*c.* 1960). Effectively, this eclectic approach is still dominant, and what this means in practice is that virtually all recent work can be located somewhere upon a spectrum from the more internalist [e.g. 76; 86; 221] shading through to the more externalist [e.g. 225; 169]. But the new eclectics, unlike externalists, recognize that scientific judgements about pertinent experimental or analytical results, or about correct theory, can sometimes only be understood in terms of the technical tradition within which they play a part, and may be insulated from wider social considerations. This is not tantamount to internalism, however, since eclectic historians of science would argue (or assume) that in such cases the technical tradition itself is a socially constructed, or culturally determined, phenomenon and that work within that tradition is affected by social interactions between the relevant specialists [277: 352–3; 271: 222; 273; 101; 68; 291].

The important thing to note about the historiography of science, then, is that ever richer contextualization has been the main ambition of the majority of its practitioners for a number of decades. The result is a sub-discipline of history which is flourishing in its own terms, and which, more generally, is making a major contribution to our understanding of how and why science has become such an overwhelming feature of Western culture. Within this general effort to understand the cultural dominance of science, accounts of the Scientific Revolution play a major role. But it is important, whenever we consider these accounts, to bear in mind the complex historiographical issues which have helped to shape them.

Another important aspect of the historiography of science, and its striving for increased contextualization, is the increasing concern with the practice of science as a day-to-day activity. It seems fair to say that the earliest examinations of Western science as a phenomenon in its own right were undertaken by philosophers. The history of science, in its early development as a sub-discipline of history, was often used to provide case histories to illustrate the claims of philosophers of science about the scientific method, or the way new discoveries were made, or how they were established as 'true'. The result of this philosophical orientation was an almost universal concern with scientific thinking and scientific ideas.

It was not long, however, before the history of science began to offer counter-examples to philosophical claims and to propose its own alternative accounts of science and its development. The classic case of the history of science rejecting previous philosophical theorizing about science was Thomas Kuhn's *Structure of Scientific Revolutions* [187]. Kuhn's historically based theory about how scientific knowledge can be said to advance remains highly influential, but chiefly (and somewhat ironically) among *philosophers* of science who continue to debate whether Kuhn was correct. Historians have always preferred to stick to historical research, rather than to debate what a previous historian (much less a philosopher) has said, and so Kuhn's great work has proved less influential in historiography, as historians of science have continued to scrutinize in ever finer detail every episode of scientific discovery, or advance, and thereby have demonstrated that every case is unique.

If the historiography of science has tended to turn away from the philosophy of science, it has recently found a more amenable ally in the sociology of science. One major aspect of this new alliance is a shift from the intellectual history of science, in which the development of science is seen as a history of successive ideas

about the way the world works or a history of scientific thinking, to a social history of science, in which the focus is on the role and function of the student of nature in the wider society of which he (and later she) is a part [12; 15; 18; 25; 59; 97; 127; 138; 171; 177; 200; 201; 208; 223; 231: Part II; 267; 278; 279; 287; 289; 332]. Not all this work, it has to be said, offers much that is useful for understanding the nature of science se. Much recent work along these lines merely tends to confirm what other social historians have claimed about the nature and arrangement of earlier societies – showing that those engaged in studying nature, no less than their contemporaries, engaged with and helped to constitute the societies of which they were a part. The effect is to blend scientists in with the rest of society, rather than to try to understand anything about the institution of science. For our purposes, however, the best work on the social history of ‘science’ is that which shows how the socially determined aspects of particular practitioners’ work are relevant to our understanding of how scientific knowledge advanced, or how science developed.

Specifically, a number of socially aware studies of various technical arts and sciences have shown how the *practice* of those arts and sciences is crucial to our understanding of how they contributed to contemporary developments, and which ultimately led to the formation of something recognizably like modern science. It is thanks to this work that we can now see that the history of science cannot be told as merely a history of great ideas, a succession of brilliant intellectual insights. If we wish to approach a full understanding, we need to recognize that the way in which various practitioners did their work was as important a factor in the development of modern science as the more abstract philosophical thinking which is all too often presented in intellectual histories of astronomy, mathematics or physics, or in the presentations of Bacon, Galileo, Boyle and Descartes as ‘great thinkers’ [which include, for example, items 2, 3, 82, 90, 185, 228, 229, and 246 in the Bibliography; items which should not be discarded, but need to be complemented with more recent studies]. These issues will be addressed in the appropriate places throughout the following pages, but it is perhaps worth noting a few examples here.

Some of the most important recent work in the history of science has made clear just how distinct and separate mathematical practitioners were from the university professors who studied and taught natural philosophy. This was not just true of those mathematical practitioners who worked outside the university system, as surveyors, military engineers and so on, but even those who worked within the universities, such as teachers of astronomy

and astrology [321; 18; 15; 79]. A great deal of recent work in the history of science has shown how the pragmatic value of mathematical studies began to be recognized in the Renaissance period, and this in turn meant that mathematicians increased in social status, eventually to the point where they could be seen to be as worthy of note as the formerly exclusively elite natural philosophers [18; 319]. The upshot was that natural philosophy itself, once entirely contemplative and qualitative, became increasingly pragmatic and quantitative as mathematicians showed the relevance of their work to an understanding of nature. In this historical process, natural philosophy itself was completely transformed; but that transformation cannot be understood solely in terms of new ideas, to a large extent it was a change of practice, a change in how natural philosophy was conducted. The summation of this, perhaps, can be seen in Isaac Newton's *Mathematical Principles of Natural Philosophy*. Even the title can be seen as a clarion call, alerting contemporary educated readers to the fact that the author is demonstrating that, contrary to usual expectations about the separate standing of natural philosophy and mathematics, natural philosophy is based on mathematical principles. But, it can also be seen that by doing things the way mathematicians do, Newton developed a different natural philosophy – one in which it is possible to talk meaningfully (that is, mathematically) about a force capable of operating over vast distances of empty space (the force of gravity), even though traditional natural philosophers would have declared all talk of actions at a distance to be nonsensical [145; 146].

Similarly, recent work has shown the importance of alchemy in the transformation of traditional natural philosophy into something closer to modern science. Again, the issue is not just one of introducing ideas derived from alchemy into natural philosophy, but of insisting that alchemical procedures provide the only certain means of discovering the nature of matter and, therefore, the only proper foundation for an understanding of the material world. The result was a new theory of matter, established by alchemical practice, which necessarily implied a new kind of natural philosophy. As was the case in mathematics, developments along these lines were the result of contributions from many practitioners, but they can be seen to culminate in the revisionist assumptions of many leading alchemically inclined natural philosophers, including Francis Bacon [247; 248; 117; 150], Robert Boyle [238; 239; 220; 40] and, once again, Isaac Newton, who was as much an alchemist as he was a mathematician [320; 77; 78].

It has even been suggested that numerous artisans and craftsmen, by doing things their way, led others to a new, more

practice-led way of doing natural philosophy. In particular, the experimental method, one of the characterizing features of the Scientific Revolution, has been claimed to have derived from various craft practices, although it is by no means always clear how routine practical techniques are meant to have inspired experimental investigations [331; 332; 287; 12]. It is easy to see, however, that the attempts of Renaissance artists to develop geometrical perspective, as a technique of representing the three-dimensional world in two dimensions, might have bolstered the attempts of mathematicians to persuade natural philosophers that mathematics could reveal aspects of the natural world [96; 222]. Similarly, the artists' striving for realism in depictions of plants and animals may have had an influence on the attitudes of those concerned with natural history [222; 7].

As a result of still growing historical research into these areas, to say nothing of research on the role of medical practitioners [25; 29; 30; 49; 51; 71; 74; 251; 307; 311, 414], and practitioners of what was known as 'natural magic' [53; 54; 122; 136; 139; 146; 150; 153, 252; 312], it is now clear that the history of science is not just a history of theoretical developments, but also a history of developing techniques and practices. It should be obvious that this must be so. After all, modern science depends as much upon experimental and other specialist techniques, as it does upon abstract theorizing. Indeed, so important are such practical techniques and procedures taken to be, that the speculations of theorists often have to await confirmation from experimentalists before they are judged to be sound. If our concern is with understanding the nature of modern science by looking at its history, then we must look to see where this fruitful alliance between theory and practice originated. As with so many other aspects of science, it can be seen to have originated in the Scientific Revolution.